

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT :- CYBER SECURITY
PRCATICAL NO: 4

AIM:- Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

CODE:

```
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.asymmetric import rsa, padding

# Step 1: Generate RSA private key
private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

public_key = private_key.public_key()

message = input("Enter the message: ").encode()

signature = private_key.sign(
    message,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)

print("\nDigital Signature Generated Successfully!")

verify_message = input("\nEnter message to verify: ").encode()

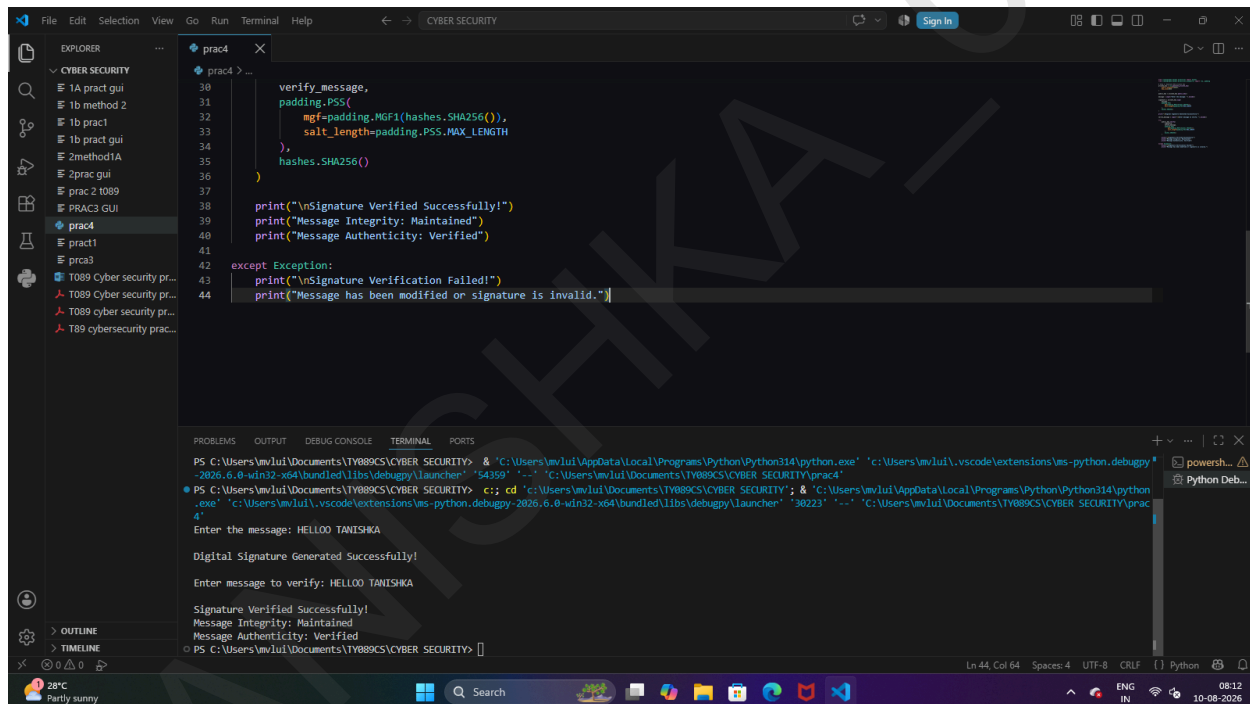
try:
    public_key.verify(
        signature,
        verify_message,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
    ),
```

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```
hashes.SHA256()  
  
)  
  
print("\nSignature Verified Successfully!")  
print("Message Integrity: Maintained")  
print("Message Authenticity: Verified")  
  
except Exception:  
    print("\nSignature Verification Failed!")  
    print("Message has been modified or signature is invalid.")
```

OUTPUT:



```
File Edit Selection View Go Run Terminal Help  
CYBER SECURITY  
prac4  
30  
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except Exception:  
    print("\nSignature Verification Failed!")  
    print("Message has been modified or signature is invalid.")  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\mv\luj\Documents\TY089CS\CYBER SECURITY> & "C:\Users\mv\luj\AppData\Local\Programs\Python\Python314\python.exe" "c:\Users\mv\luj\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher" "54359" "--" "C:\Users\mv\luj\Documents\TY089CS\CYBER SECURITY\prac4"  
PS C:\Users\mv\luj\Documents\TY089CS\CYBER SECURITY> c:\Users\mv\luj\Documents\TY089CS\CYBER SECURITY> cd "C:\Users\mv\luj\AppData\Local\Programs\Python\Python314\python.exe" "c:\Users\mv\luj\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher" "38223" "--" "C:\Users\mv\luj\Documents\TY089CS\CYBER SECURITY\prac4"  
Enter the message: HELLOO TANISHKA  
Digital Signature Generated Successfully!  
Enter message to verify: HELLOO TANISHKA  
Signature Verified Successfully!  
Message Integrity: Maintained  
Message Authenticity: Verified  
PS C:\Users\mv\luj\Documents\TY089CS\CYBER SECURITY>
```

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GUI:

```
import tkinter as tk
from tkinter import messagebox
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.asymmetric import rsa, padding

private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

public_key = private_key.public_key()

signature = None

def sign_message():
    global signature

    message = message_entry.get("1.0", tk.END).strip()

    if message == "":
        messagebox.showwarning("Warning", "Please enter a message.")
        return

    signature = private_key.sign(
        message.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )

    status_label.config(
        text="Digital Signature Generated Successfully!",
        fg="green"
    )
```

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```
signature_text.delete("1.0", tk.END)
signature_text.insert(
    tk.END,
    signature.hex()
)

def verify_message():
    if signature is None:
        messagebox.showwarning(
            "Warning",
            "Please generate the digital signature first."
        )
        return

    message = verify_entry.get("1.0", tk.END).strip()

    if message == "":
        messagebox.showwarning(
            "Warning",
            "Please enter a message to verify."
        )
        return

    try:
        public_key.verify(
            signature,
            message.encode(),
            padding.PSS(
                mgf=padding.MGF1(hashes.SHA256()),
                salt_length=padding.PSS.MAX_LENGTH
            ),
            hashes.SHA256()
        )

        result_label.config(
            text="✓ Signature Verified Successfully!\n"
                "✓ Message Integrity Maintained\n"
                "✓ Message Authenticity Verified",
            fg="green"
        )
```

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```
)

except Exception:
    result_label.config(
        text="X Signature Verification Failed!\n"
        "Message has been modified or signature is invalid.",
        fg="red"
    )

def clear_all():
    message_entry.delete("1.0", tk.END)
    verify_entry.delete("1.0", tk.END)
    signature_text.delete("1.0", tk.END)

    status_label.config(text="")
    result_label.config(text="")

root = tk.Tk()
root.title("RSA Digital Signature")
root.geometry("700x700")
root.resizable(False, False)

title = tk.Label(
    root,
    text="RSA DIGITAL SIGNATURE",
    font=("Arial", 20, "bold")
)
title.pack(pady=15)

tk.Label(
    root,
    text="Enter Message:",
    font=("Arial", 12, "bold")
).pack()

message_entry = tk.Text(
    root,
    height=4,
    width=70
```

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```
)  
message_entry.pack(pady=8)  
  
tk.Button(  
    root,  
    text="Generate Digital Signature",  
    command=sign_message,  
    font=("Arial", 11, "bold"),  
    padx=10,  
    pady=5  
) .pack(pady=5)  
  
status_label = tk.Label(  
    root,  
    text="",  
    font=("Arial", 11, "bold")  
)  
status_label.pack(pady=5)  
  
tk.Label(  
    root,  
    text="Digital Signature:",  
    font=("Arial", 12, "bold")  
) .pack()  
  
signature_text = tk.Text(  
    root,  
    height=5,  
    width=70  
)  
signature_text.pack(pady=8)  
  
tk.Label(  
    root,  
    text="Enter Message to Verify:",  
    font=("Arial", 12, "bold")  
) .pack()  
  
verify_entry = tk.Text(  

```

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```
        root,
        height=4,
        width=70
    )
verify_entry.pack(pady=8)

tk.Button(
    root,
    text="Verify Signature",
    command=verify_message,
    font=("Arial", 11, "bold"),
    padx=20,
    pady=5
).pack(pady=5)

result_label = tk.Label(
    root,
    text="",
    font=("Arial", 11, "bold"),
    justify="center"
)
result_label.pack(pady=15)

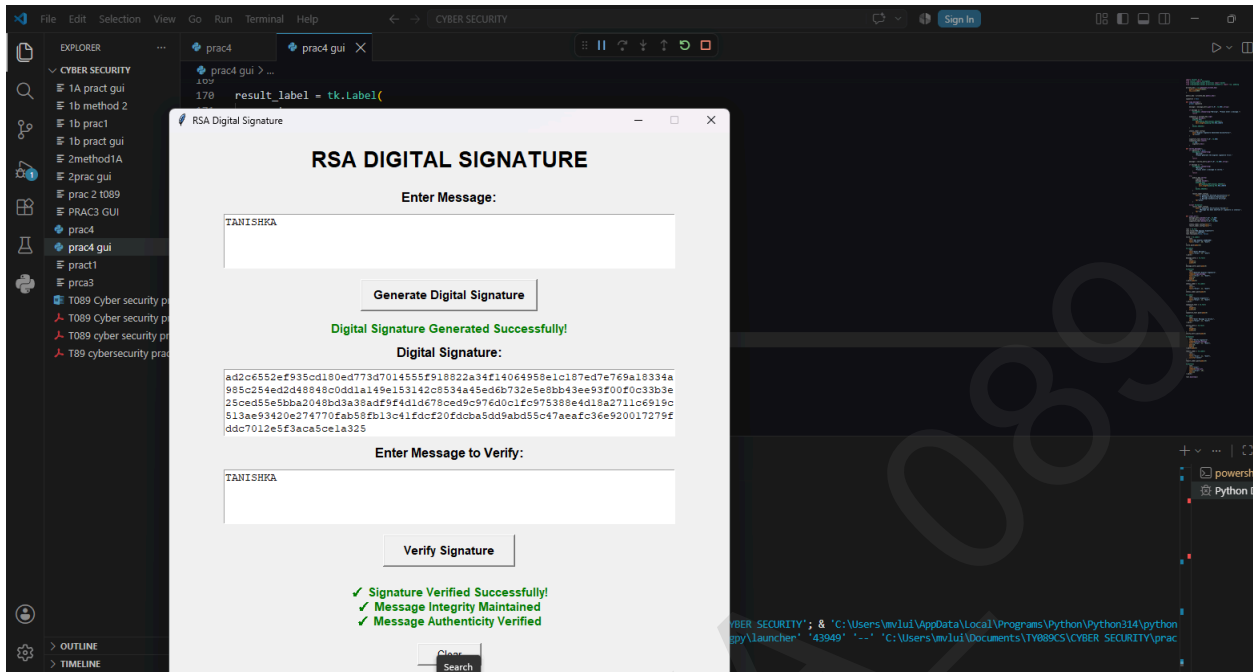
tk.Button(
    root,
    text="Clear",
    command=clear_all,
    font=("Arial", 10),
    padx=20
).pack()

root.mainloop()
```

OUTPUT:-

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